

Manual

There are 3 python files in the codebase. The main tool is the 'NeuralNetwork.py'. The baseline to compare is 'LinearBaseline.py' and the 'comparison.py' is a script that is used to display a comparison between the results and run the statistical test.

When 'NeuralNetwork.py' is run, it opens a command line interface with 4 options.

1. Start training - this option begins to create neural network models if they don't already exist and begins training them for the given systems in /datasets. This options also tests the models and prints out the results. The results are also written to a JSON-like txt file 'neural_network_results.txt'.
2. Start testing - this options only tests the models and prints out the results. The results are also written to a JSON-like txt file 'neural_network_results.txt'.
3. Change systems - this option allows the user to remove or add systems to be trained. Note if the system added doesn't exist in /datasets, no model will be created.
4. Exit - exits the program.

The 'LinearBaseline.py' runs the linear regression tool on all the datasets, prints and writes the results to 'linear_regression_results.txt'. Simply run the file.

The 'comparison.py' script prints out the results from the comparison and statistical test used in the experiments. It also writes everything to a readable 'comparison_results.txt'. Simply run the file.